

Suffix-Tree Example: awyawxawz

1 Text

Text

The example text is

$$T = \text{awyawxawz}.$$

The listed suffixes are:

Position	Suffix
1	awyawxawz
2	wyawxawz
3	yawxawz
4	awxawz
5	wxawz
6	xawz
7	awz
8	wz
9	z

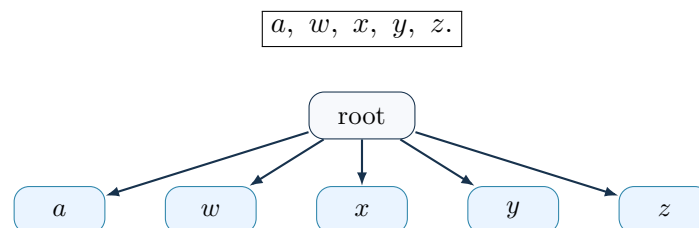
The suffix list is often written from shortest to longest, but position labels are easiest to understand in left-to-right text order.

2 Root Branches

The suffixes begin with

$a, w, y, x, z.$

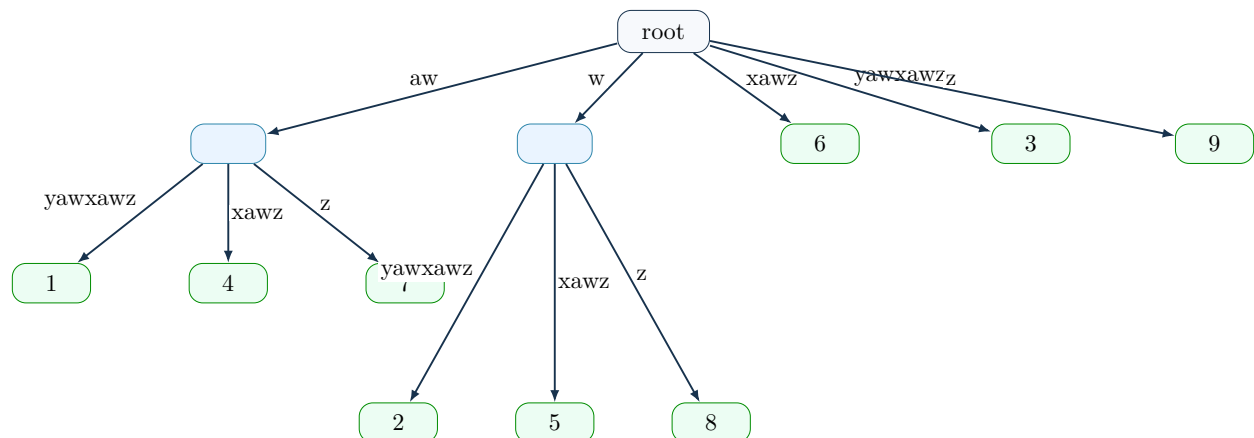
Therefore the root has branches:



3 Grouped Suffixes

Branch	Suffixes in that group, with starting positions
a	1 : awyawxawz, 4 : awxawz, 7 : awz
w	2 : wyawxawz, 5 : wxawz, 8 : wz
x	6 : xawz
y	3 : yawxawz
z	9 : z

4 Full Suffix Tree



How to read the tree

Each leaf stores a starting position. Read the edge labels from the root to a leaf to recover the suffix at that position.

5 Stored Positions

Suffix-tree leaves can store the starting positions of suffixes in the original text. For

$$T = \text{awyawxawz},$$

the suffix beginning at position 1 is awyawxawz, and the suffix beginning at position 9 is z.

Search Use

When a pattern is matched, the leaves below the matched point identify all starting positions where that pattern occurs.

6 Example Pattern

The pattern

$$P = \text{aw}$$

matches suffixes beginning with aw:

$$\text{awz}, \quad \text{awxawz}, \quad \text{awyawxawz}.$$

So the pattern occurs at the corresponding starting positions.

$$P = \text{aw} \text{ occurs at positions } 1, 4, 7.$$

7 Another Pattern

For

$$P = \text{wx},$$

search begins at the w -branch. Among the w -suffixes,

5 : wxawz

starts with wx. Hence

wx occurs at position 5.
